On-Demand Calculations — LST & Sky View Factor

Actions menu → Calculate for this view: the current map view is computed in numpy and stored as one raster row

BROWSER

LstTool.js #lst-card
lists Landsat scenes
over the view
(ranked by clear share)
Calculate → POST bbox

Actions menu
Calculate for this view
opening one card
closes the other

SvfTool.js #svf-card
asks /svf/info/ on open
and after each map move
radius 25–200 m
cell DSM / 1 / 2 / 5 m

?bbox

LST · URBAN_HEAT/LST/ · RS_IMAGERY, LANDCOVER → LST

lst/views.py
GET /urban_heat/lst/scenes/
POST /urban_heat/lst/calculate/
(calculate: staff only)

pipeline.calculate_lst()
NDVI layers in the view,
merged newest first
30 m grid, EPSG:28992
≤ 48 MB download (~40×40 km)

gee.py · Earth Engine
Landsat 8/9 B10 TOA
+ L2 QA_PIXEL · ASTER GED
TPW: NCEP, then ERA5
(NCEP stops 2026-03-18)

smw.py (numpy)
NDVI → FVC → emissivity
Statistical Mono-Window
(Ermida et al.) → LST °C
inputs only from GEE

Grid = the view clipped to the NDVI layers' coverage. A scene with no water-vapour source for its day fails before the download. The TPW source is kept in measurement_method.

SVF · PHYSICALENV/SVF.PY · DSM → SVF

?bbox

physicalEnv/views.py
GET /physicalEnv/svf/info/
POST /physicalEnv/svf/calculate/
(calculate: staff only)

calculate_svf() · plan()
dsm_for_view(): DSM covering
the view + a halo of one radius
refused above 16 M cells
→ Recalculate if stored

load_dsm()
window of dsm_raster, or
the DSM's COG warped back
to the storage CRS
at the COG's own cell

sky_view_factor() (numpy)
RVT horizon search
heights ÷ cell size
SVF = mean(1 − sin h)
halo cropped off

Reads only the view plus the halo, so the horizon at the edge of the view is still seen. One SkyViewFactor row per view; source = "DSM <id>: …".

LST grid

JSON: new + replaced layers

SVF grid

Map UI update
Layers.removeLayer(replaced)
+ raster_layer_entry(new)

post_save → COG
core/signals.py
export_raster_to_cog() → TiTiler

save_lst() / save_svf() · replace_covered()
one raster row per view: LandSurfaceTemperature (°C) / SkyViewFactor
a recalculation replaces the covered rows of the same scene / DSM (+ COG files)
no receiver: a post_save must never start an Earth Engine download

Both are synchronous requests (LST ~10–60 s). Earth Engine only serves the inputs; every equation runs in numpy.
Same tools from the shell: manage.py calculate_lst --city Enschede · physicalEnv.svf.calculate_svf(bbox=…).

Browser · Django view / orchestration · Earth Engine · numpy / raster · Storage